

ASSESS AND MANAGE TB INFECTION

TB tests changing from 'Xpert Ultra' to 'TB NAAT'
(NAAT = nucleic acid amplification test and includes Xpert as well as newer TB tests).

'TB infection' is different from 'TB disease'. TB infection refers to TB bacteria that has entered the body but is not yet making the body sick – often called latent TB, which means hidden/inactive TB.

Assess the need for TB preventive treatment (TPT)

Is patient a TB contact: has s/he shared an enclosed space at work, socially, or in a household, for ≥ 1 night or for frequent/extended daytime periods, with an adult/adolescent with lung TB ("index patient")?

Yes

Did patient share this space during the 3-month period before the index patient started their TB treatment?

Yes

- Check for active TB disease: send 1 sputum sample for TB NAAT, regardless of symptoms → 92. Where needed, arrange CXR and do urine LAM test → 92.
- Assess clinically: if TB symptoms¹, discuss/refer for investigation.

If no TB symptoms, clinically well and investigations negative², active TB disease unlikely. Give TPT. If patient has had previous TPT, repeat course with every new exposure.

Choose what TPT to give: start by checking drug sensitivity results of index patient³:

Resistance to rifampicin or INH

- If resistance to INH only, give **4R**.
- If resistance to rifampicin, arrange chest x-ray and doctor review:
 - If chest x-ray normal, check index patient's INH resistance result:
 - If *no resistance* to INH on phenotypic DST: give **6H** → 90.
 - If resistance to INH (or unknown): check fluoroquinolone resistance and discuss with TB specialist/refer.
 - If chest x-ray abnormal: avoid giving TPT. Send 1 sputum sample for TB NAAT, TB microscopy, culture and DST. Refer to TB specialist same week.

Susceptible to rifampicin or INH (or unknown)

Choose TPT regimen according to age and HIV status:

Child < 25kg

If newborn exposed to TB, or newborn tests positive for HIV, assess and manage → 167.

HIV exposed infant

Infant on ART

Infant on PEP

Infant no longer on PEP

Give **6H** → 90.

HIV unexposed infant

Give **3RH** or **6H** → 90.

No

- HIV positive
- Known with silicosis⁴

Any of:

- Known immunocompromise or taking immunosuppressive medications⁵

One or more of above

- Check for active TB disease: send 1 sputum sample for TB NAAT, regardless of symptoms → 92. Where needed, arrange CXR and do urine LAM test → 92.
- Assess clinically: if TB symptoms¹, discuss/refer for investigation.

If no TB symptoms, clinically well and investigations negative², active TB disease unlikely. If patient has had TPT previously, no need to repeat TPT. If no previous TPT, give TPT.

No

None of above

TB preventive treatment (TPT) is *not* needed.

Continue routine care.

6H – 6 months isoniazid; **12H** – 12 months isoniazid; **3RH** – 3 months rifampicin and isoniazid; **3HP** – 3 months isoniazid and rifapentine; **4R** – 4 months rifampicin

¹TB symptoms in adults may include: current cough, weight loss, drenching night sweats, fever or coughing up blood. TB symptoms in children may include: current cough, poor weight gain/failure to thrive, fever, lethargy or decreased playfulness, visible neck mass. ²If investigations are not available, continue to give TPT if patient has no symptoms of TB. ³If drug susceptibility results of index patient unknown, ask where index patient receives TB treatment and contact clinic for treatment details. ⁴Silicosis is a chronic lung disease caused by breathing in silica dust while working in mining or construction. ⁵Cancer, those waiting/received blood/organ transplant or receiving chemotherapy, dialysis or corticosteroids, diabetes). ⁶Cancer, uncontrolled diabetes, those awaiting/received blood/organ transplant or receiving chemotherapy, dialysis or long term corticosteroids.